

CubeDesign 2026

CanSat Category

1. General Objective of the Competition

CubeDesign CanSat 2026 challenges teams to design, integrate, document, test, and operate a CanSat-type system capable of executing a descent mission, acquiring data, and transmitting information during a flight profile simulated through drone deployment. The competition aims to introduce participants to fundamental space engineering concepts, including system architecture, telemetry, mission control, mechanical design, and experimental validation.

2. General Regulations

- Each team must consist of **2 to 10 students** plus **1 supervising professor**.
- Schools, institutes, or interdisciplinary groups may participate.
- The system must comply with the dimensional and mass constraints defined for CanSat:
 - Mass: $500\text{ g} \pm 100\text{ g}$ (excluding payload)
 - Geometry compatible with the envelope defined in this regulation
- Activities are organized in order of complexity, not necessarily chronological.
- The competition consists of two stages:
 - Documentation evaluation
 - On-site testing
- Documentation must be submitted in English.

3. Competition Stages

The competition consists of two stages:

3.1 Stage 1 – Documentation Evaluation

Two months prior to the on-site tests, each team must submit a Design Package (DP) including:

- Mission concept
- System architecture
- Mass, power, and data budget
- Payload design
- Verification and Validation Plan (V&V)
- Engineering model description:
 - Block diagram
 - Electrical architecture
 - Communication system
 - Algorithms

- Testing procedures

Only selected teams will advance to the on-site stage.

3.2 Stage 2 – On-site Testing

To be held at UNSAM, including:

- Mission execution
- Technical tests (basic and intermediate)
- Descent and landing evaluation
- Final presentation

4. Technical Tests

Tests are organized in increasing levels of complexity.

4.1 Basic Level

- Command reception (TC)
- Telemetry transmission (TM)
- Flight data recording

4.2 Intermediate Level

- Landing system activation:
 - Autonomous
 - Telecommanded
 - Pre-triggered
- Altitude measurement during descent
- Data logging and storage

4.3 Mission

The CanSat 2026 mission simulates the behavior of a system released from an aerial platform, descending using a passive or active system, and acquiring environmental data during descent. The CanSat will be deployed from a drone (approximately 100 m).

The system must be capable of:

- Operating in a “closed” state during launch
- Being released from a drone
- Deploying its descent system
- Measuring altitude during descent
- Recording and transmitting data:
 - Acceleration
 - Altitude
 - Velocity
 - Ambient temperature
 - Ambient humidity

- Atmospheric pressure
- Surviving ground impact

4.3.1 Mission Objectives

Primary Objective

Design a CanSat system capable of:

- Executing a controlled descent
- Measuring altitude variation
- Recording data
- Transmitting information to the ground segment

Secondary Objectives

- Evaluate design decisions
- Analyze landing system performance
- Validate data acquisition strategies

4.3.2 General System Architecture

Each team must implement:

- CanSat structure
- Power system
- On-board computer
- Communication system
- Payload:
 - Sensors
- Landing system

5. Software and Data Management

The system must:

- Manage operational modes (launch, descent, post-impact)
- Record mission data
- Transmit structured telemetry

Data must be:

- Reproducible
- Documented
- Suitable for post-analysis

6. Verification and Validation (V&V)

Each team must submit a V&V plan including:

- Verification methods
- Requirement-to-test traceability
- Functional testing

Not all tests will be executed during the event.

7. Mechanical Envelope

The mechanical envelope must be a cylinder of 120 mm height and 30 mm radius. The lower face (impact side) must contain the payload. The maximum system mass (excluding payload) must be $500\text{ g} \pm 100\text{ g}$.

8. Evaluation

8.1 Technical Evaluation

- Fit-check (eliminary)
- Mass (eliminary)
- Telemetry
- Altitude measurement
- Mission execution

8.2 Landing system

- Autonomous
- Telecommanded
- Pre-triggered

8.5 Mass

- System weight optimization

8.6 Project

- Technical coherence
- Level of documentation

9. Final Presentation

Each team must present:

- System design
- Mission results
- Altitude data
- Performance analysis

Evaluated by a technical jury.

10. General Observations

- The landing system must be contained within the CanSat
- The CanSat structure must not show observable structural damage after landing
- Teams must bring their own ground station
- Launch will be subject to weather conditions
- The organization reserves the right to modify these regulations to minimize launch risks
- Clarifications will be published prior to the event

High Level Requirements

CanSat Mission – CubeDesign 2026

1. General System Requirements

HLR-GEN-01

The system shall fit within a cylindrical envelope of 120 mm height and 60 mm diameter, and shall have a total mass between 400 g and 600 g, excluding payload.

- Mass: 500 g $\pm$ 100 g
- Shape: Cylindrical
- Height: 120 mm
- Radius: 30 mm

HLR-GEN-02

The system shall be composed of the following identifiable subsystems: structure, power system, processing unit, communication system, payload, and descent system.

HLR-GEN-03

The system shall operate autonomously during all mission phases, while allowing external interaction through telecommands and telemetry exchange.

HLR-GEN-04

The system architecture shall be documented and shall include justification of design decisions, including subsystem selection and interfaces.

2. Mission Requirements

HLR-MIS-01

The system shall withstand deployment from a drone at an altitude of approximately 100 m while in a fully enclosed configuration, without loss of functionality.

HLR-MIS-02

The system shall initiate and sustain a controlled descent using an internal descent mechanism.

HLR-MIS-03

The system shall measure altitude during descent with a sampling rate of at least 1 Hz and store the data for post-mission analysis.

HLR-MIS-04

The system shall acquire environmental data during descent, including at minimum temperature, humidity, and pressure, with time-stamped measurements.

HLR-MIS-05

The system shall survive ground impact and retain the ability to store or transmit mission data after landing.

3. Communication Requirements**HLR-COMM-01**

The system shall transmit telemetry data to the ground segment during descent and for at least 30 seconds after landing.

HLR-COMM-02

The system shall be capable of receiving at least one valid telecommand during the mission or execute a predefined autonomous sequence.

HLR-COMM-03

La arquitectura de comunicaciones (frecuencia, protocolo, tasa de datos) deberá ser definida y justificada por el equipo.

HLR-COMM-04

The system shall operate with an equivalent isotropically radiated power ≤ 30 mW and within locally permitted frequency bands.

HLR-COMM-05

Each team shall provide a functional ground station capable of receiving telemetry and issuing commands.

4. Payload Requirements**HLR-PL-01**

The payload shall include environmental sensors capable of measuring temperature, humidity, and pressure.

HLR-PL-02

The payload shall be located and mechanically protected such that it remains operational after ground impact.

5. Landing Requirements**HLR-LAND-01**

The system shall implement a landing mechanism that reduces the impact velocity relative to free fall.

HLR-LAND-02

The landing mechanism shall be one of the following types: autonomous, telecommanded, or pre-triggered.

6. Software and Data Requirements

HLR-SW-01

The system shall implement at least three operational modes: pre-launch, descent, and post-impact.

HLR-SW-02

The system shall record mission data in a structured format suitable for post-processing and analysis.

7. Verification and Validation Requirements**HLR-VV-01**

The system shall include a verification plan demonstrating compliance with all applicable high-level requirements.

HLR-VV-02

Verification methods shall include at least one of the following: test, analysis, or demonstration.

8. System Coherence Requirements**HLR-SYS-01**

The system design, mission objectives, verification activities, and obtained results shall be consistent and traceable within the system documentation.